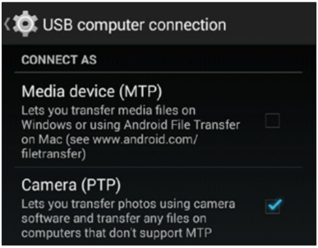

Figure 1: USB modes of an Android device

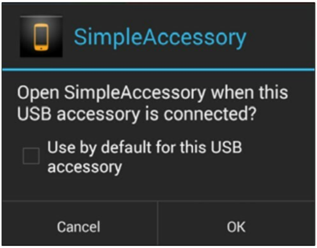

Figure 2: Launching an application after initialisation

- BeagleBone powered with TI Starterware
- And currently, any hardware with USB host capabilities powered by Linux.

So we decided to add external hardware as the ADK host to an Android device. Let's see how ADK can initialise the Android device in the accessory mode.

Initialising the Android device

Once the Android device is connected to the ADK host, the device will enumerate with its original configuration, which may vary for each device. Initially, the device may be connected in the camera (PTP support) or media device (MTP support) mode to the host. Now, ADK should initialise the device to turn into the accessory mode. This can be done in the following few steps by sending various vendor (Google) specific control requests through the control endpoint whose address is zero.

Step 1: Query the protocol using request code 51(0x33), which returns 1 for AOA and 1.0 or 2 for AOA 2.0; other attributes as per USB specifications are listed here.

bmRequest type	TYPE_VENDOR | DIR_DEVICE_TO_HOST		
bRequest	51		
wValue	0	wIndex	0
wLength	2	Data	base_address_of_two_bye_buffer

Step 2: Send identity strings using request code 52(0x34) with the suitable value for each string ID.

bmRequest type	TYPE_VENDOR | DIR_HOST_TO_DEVICE		
bRequest	52		
wValue	0	wIndex	string id
wLength	Length_of_string	Data	base_addr_of_string

Here, 'string id' is a value between 0 to 5, which represents each type of identity string as listed below.

manufacturer	0	model	1	Description	2
version	3	url	4	Serial id	5

Step 3: The final step is to start the device in the accessory mode using the request code 53(0x35) as follows:

bmRequest type	TYPE_VENDOR | DIR_HOST_TO_DEVICE		
bRequest	53		
wValue	0	wIndex	0
wLength	0	Data	NULL

At this point, if essential identity strings (manufacturer, model) are not sent, the device can start in the accessory mode without associating with any application as allowed in AOA 2.0.This is useful to start the device just in audio or HID mode.

ADK can act as an audio accessory as supported in AOA 2.0, where audio output from the Android device is streamed to the ADK host without the need of any application in two-channel 16-bit PCM format, i.e., the Android device can be identified as a capture device by an audio sub-system like ALSA on the Linux side. So if you are planning to use ADK as an audio dock for your Android device, you can send another request with the following attributes, and if you do, this must be sent before starting in the accessory mode (53) as per AOA specs.

```
bRequest =58(0x3A), wValue=1
```

The rest is the same as above.

Also, the ADK host can act as an HID event generator or HID proxy as supported in AOA 2.0, for that the following request codes are available:

```
54(0x36)   register ADK as HID source     55(0x37)
unregister the ADK
56(0x38)   set hid report descriptor     57(0x39)   send a
hid event
```